PROGRAMMING IN JAVA

Assignment XII

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which JDBC object is used to iterate over records returned by a SELECT query?

a. `Statement`

b. `Connection`

c. `ResultSet`

d. `DriverManager`

Correct Answer: c

c. ResultSet

Detailed Solution:

A `ResultSet` object stores and allows traversal of rows returned by a SELECT query.

QUESTION 2:

Which method is used to display a Swing `JFrame` on the screen?

a. `display()`

b. `setVisible(true)`

c. `showFrame()`

d. `open()`

Correct Answer: b

b. setVisible(true)

Detailed Solution:

A Swing frame becomes visible only after calling `setVisible(true)`.

QUESTION 3:

What is the index of the first element in a Java array?

a. **1**

b. **-1**

c. **0**

d. **Depends on array size**

Correct Answer: c

c. **0**

Detailed Solution:

Java arrays are zero-indexed.

QUESTION 4:

Consider the following program: (note space appears only one at a time)

```
public class Question {
    public static void main(String[] args) {

        String str = "NPTEL - Programming in JAVA - JAN 2026";

        System.out.println(str.length());
    }
}
```

a. 38

b. 39

c. 40

d. 41

Correct Answer: a

a. 38

Detailed Solution:

The provided Java program calculates and prints the length of the string `str`, which contains the text "NPTEL - Programming in JAVA - JAN 2026".
The output of this program will be: 38

QUESTION 5:

Which keyword is used to handle exceptions in Java?

a. `error`

b. `handle`

c. `catch`

d. `throwable`

Correct Answer: c

c. catch

Detailed Solution:

The `catch` block handles exceptions thrown in the `try` block.

QUESTION 6:

What is the output of the following program?

```
public class Test {
    public static void aMethod() throws Exception {
        try {
            throw new Exception();
        } finally {
            System.out.print("finally ");
        }
    }

    public static void main(String args[]) {
        try {
            aMethod();
        } catch (Exception e) {
            System.out.print("exception ");
        }
        System.out.print("finished ");
    }
}
```

a. **finally**

b. **exception finished**

c. **finally exception finished**

d. **Compilation fails**

Correct Answer: c

c. **finally exception finished**

Detailed Solution:

The program is syntactically correct and here for two try blocks, there is one catch block.
Here's the step-by-step explanation:

I. The `main` method calls `aMethod()`, which throws an `Exception`.
II. Inside `aMethod()`, an `Exception` is thrown in the `try` block.
III. The `finally` block is always executed, regardless of whether an exception is thrown or not. In this case, it prints "finally ".
IV. Since `aMethod()` throws an `Exception`, control moves to the `catch` block in `main`.
V. The `catch` block prints "exception ".
VI. After the `try-catch` block in `main`, "finished " is printed.
VII. Therefore, the complete output is "finally exception finished".

QUESTION 7:

What is the output of the following program?

```
class Program {
    public static void main(String[] args) {
        int counter = 10;
        do {
            System.out.print(2 / counter);
            counter--;
        } while (counter > 2);
    }
}
```

a. **00000012**

b. **00000000**

c. **10011001**

d. **12211221**

Correct Answer: b

b. **00000000**

Detailed Solution:

The provided program uses a do-while loop to iterate from counter = 10 down to counter = 3. During each iteration, it prints the result of 2 / counter.

1. counter = 10: 2 / 10 results in 0 (integer division).
2. counter = 9: 2 / 9 results in 0.
3. counter = 8: 2 / 8 results in 0.
4. counter = 7: 2 / 7 results in 0.
5. counter = 6: 2 / 6 results in 0.
6. counter = 5: 2 / 5 results in 0.
7. counter = 4: 2 / 4 results in 0.
8. When counter = 3: 2 / 3 results in 0.

After counter becomes 2, the loop condition counter > 2 fails, and the loop terminates.
Therefore, the output of the program is 00000000.

QUESTION 8:

What should be the value of X and Y for the output of the below program to be 36?

```
public class Question {
    public static void main(String[] args) {
        int X = 4;
        int Y = 5;
        int sum = 0;
        for (int i = 0; i < X; i++) {
            for (int j = i; j < Y; j++) {
                sum = sum + j;
            }
        }
        System.out.print(sum);
    }
}
```

a. **X = 6 and Y = 5**

b. **X = 2 and Y = 7**

c. **X = 1 and Y = 10**

d. **X = 4 and Y = 5**

Correct Answer: d

d. **X = 4 and Y = 5**

Detailed Solution:

The following is the output of the program for different values of X and Y:

a. *when* X = 6 and Y = 5 *output is* 40

b. *when* X = 2 and Y = 7 *output is* 42

c. *when* X = 1 and Y = 10 *output is* 45

d. *when* X = 4 and Y = 5 *output is* 36

QUESTION 9:

Which of the following options correctly initializes the elements of the numbers array with values 1, 2, 3, 4, and 5?

```
public class NPTEL {
    public static void main(String[] args) {
        int[] numbers = new int[5];
        // #1 : Missing code block
        System.out.println("First element: " + numbers[0]);
    }
}
```

a. **numbers = {1, 2, 3, 4, 5};**

b. **for (int i = 1; i < numbers.length; i++) {**
   numbers[i] = i;
   }

c. **numbers[] = {1, 2, 3, 4, 5};**

d. **numbers = new int[]{1, 2, 3, 4, 5};**

Correct Answer: d

d. **numbers = new int[]{1, 2, 3, 4, 5};**

Detailed Solution:

`numbers = new int[]{1, 2, 3, 4, 5};` is the correct answer because it initializes the `numbers` array with values 1, 2, 3, 4, and 5 using array initializer syntax.

QUESTION 10:

Which of the following statements are correct and would NOT cause a compilation error?

```
 I.   float[] = new float(3);
II.   float f1[] = new float[];
III.  float[] f2 = new float[3];
IV.   float f3[] = new float[3];
 V.   float f4[]= { 1.0f, 2.0f, 2.0f };
VI.   float f5[] = new float[] { 1.0f, 2.0f, 3.0f};
```

a. **iii, iv, v, vi**

b. **i, ii, iii, iv**

c. **ii, iii, v, vi**

d. **i, ii, iv, vi**

Correct Answer: a

a. **iii, iv, v, vi**

Detailed Solution:

Option iii, iv, v and vi are syntactically correct for declaration of an array.